1. Power BI Fundamentals

1. What is Power BI?

Power BI is a Business Intelligence (BI) tool developed by Microsoft. It is used to connect, clean, transform, analyse and visualise data through interactive reports and dashboards.

2. What are the different components of Power BI?

The main components are:

- Power BI Desktop – Used to create reports.
- Power BI Service – Used to publish and share reports online.
- Power BI Mobile – Used to view reports on mobile devices.
- Power BI Gateway – Connects online services to local data sources.
- Power BI Report Server – Hosts Power BI reports on-premises.

3. What is the difference between Power BI Desktop and Power BI Service?

- Power BI Desktop is an application used to develop reports and perform data analysis.
- Power BI Service is an online platform used to publish, share and manage reports.

4. What is Power BI Mobile?

Power BI Mobile is an application that allows users to access and view Power BI reports and dashboards using smartphones and tablets.

5. What is a Power BI dashboard?

A dashboard is a single-page visual summary of important business information. It contains visual elements such as charts, cards and KPIs.

6. What is a Power BI report?

A report is a collection of one or more pages containing charts, tables, graphs and other visualisations used to analyse data.

7. What is the difference between a report and a dashboard?

Report	Dashboard
Can contain multiple pages	Single page
Created mainly in Power BI Desktop	Created in Power BI Service
Supports detailed analysis	Provides a quick summary
Can use multiple visuals	Uses pinned tiles

8. What are the different data sources supported by Power BI?

Power BI supports different data sources, including:

- Excel
- CSV
- SQL Server
- MySQL
- PostgreSQL
- Web
- SharePoint
- Azure
- JSON
- XML

9. What is the difference between Import Mode and DirectQuery?

- Import Mode: Data is copied and stored inside the Power BI model.
- DirectQuery: Data remains in the original database, and Power BI sends queries to retrieve the required information.

10. What is a Live Connection in Power BI?

A Live Connection connects Power BI to an existing semantic model or Analysis Services model without importing the underlying data into a new model.

11. What is a semantic model (dataset) in Power BI?

A semantic model is a structured collection of tables, relationships, calculations and other data elements used to create reports in Power BI.

Power BI Pro is a user licence that supports publishing and sharing reports. Premium capabilities provide additional capacity, scalability and advanced features. Microsoft now offers these through different licensing options, including Premium Per User and Fabric capacity.

2. Power Query and Data Transformation

13. What is Power Query in Power BI?

Power Query is a data transformation tool used to clean, filter, combine and prepare raw data before loading it into Power BI.

Example: Removing duplicate customer records from an Excel file.

14. What is the difference between Power Query and Power Pivot?

- Power Query is used to clean and transform data.
- Power Pivot is used to create data models, relationships and calculations using DAX.

15. What is the M language?

M is the programming language used by Power Query to perform data transformation operations.

Example:

```
None
  Table.SelectRows(
      Sales,
      each [Sales] > 1000

  )
```

This filters records where sales are greater than 1000.

16. What is data cleaning?

Data cleaning is the process of identifying and correcting errors, missing values, duplicate records and inconsistent data.

17. How do you remove duplicate records in Power Query?

Select the required column or columns, go to the Home tab and select Remove Rows → Remove Duplicates.

18. How do you handle missing or null values?

We can handle missing values by:

- Replacing null values with 0.
- Replacing them with the average or median.
- Removing rows containing missing values.
- Filling values using Fill Down or Fill Up.

The appropriate method depends on the dataset.

19. What is the difference between Merge Queries and Append Queries?

- Merge Queries combines tables horizontally using matching columns.
- Append Queries combines tables vertically by adding rows from one table to another.

Example: Merge combines customer information with order information using CustomerID. Append combines January sales and February sales.

20. What is the difference between a Conditional Column and a Custom Column?

- Conditional Column creates a new column using predefined IF-THEN conditions through the interface.
- Custom Column allows users to write expressions using the M language.

21. What is Query Folding?

Query Folding is the process of converting Power Query transformation steps into queries that the original data source can execute.

Example: Instead of downloading all records and filtering locally, Power BI sends the filter operation to SQL Server.

22. What is the purpose of the Applied Steps section?

Applied Steps displays all the transformations performed on the data in Power Query. It allows users to review, modify or remove individual transformation steps.

23. How do you change a column's data type?

Select the column, go to the Transform or Home tab, select Data Type and choose the required type, such as Text, Whole Number, Decimal Number or Date.

24. What is the difference between Replace Values and Fill Down?

- Replace Values replaces a particular value with another value.
- Fill Down copies the previous non-null value into empty cells below it.

25. How do you split a column into multiple columns?

Select the column, go to Split Column and choose a delimiter, number of characters or another splitting method.

Example: Splitting `Johnson  Joseph` into First Name and Last Name using a space delimiter.

26. What is the purpose of Pivot and Unpivot Columns?

- Pivot converts unique row values into separate columns.
- Unpivot converts multiple columns into attribute-value rows.

Unpivot is especially useful when preparing data for analysis.

3. Data Modelling

27. What is data modelling in Power BI?

Data modelling is the process of organising tables and creating relationships between them so that data can be analysed correctly.

Example: Connecting Sales, Customers and Products tables.

28. What is a relationship in Power BI?

A relationship is a connection between two tables using a common column.

Example: Connecting the CustomerID column in the Customers table with CustomerID in the Sales table.

29. What are the different types of relationships?

The main relationship types are:

- One-to-One (1:1)
- One-to-Many (1:*)
- Many-to-One (*:1)
- Many-to-Many (:)

30. What is a one-to-many relationship?

A one-to-many relationship means one record in the first table can be connected to multiple records in another table.

Example: One customer can place multiple orders.

31. What is a many-to-many relationship?

Answer: A many-to-many relationship occurs when multiple records in one table can be connected to multiple records in another table.

Example: Multiple students can enrol in multiple courses.

32. What is the difference between active and inactive relationships?

- Active Relationship: Used automatically when calculating data.
- Inactive Relationship: Not used automatically but can be activated in a DAX calculation using USERELATIONSHIP().

33. What is a primary key?

A primary key is a column that uniquely identifies each record in a table.

Example: CustomerID in a Customers table.

34. What is a foreign key?

A foreign key is a column that refers to the primary key of another table and helps establish relationships.

Example: CustomerID in the Sales table.

35. What is a Star Schema?

A Star Schema is a data modelling structure where one central fact table connects to multiple dimension tables.

Example: A Sales table connected to Customers, Products and Date tables.

36. What is a Snowflake Schema?

A Snowflake Schema is a data modelling structure where dimension tables are further divided into related tables to reduce repeated information.

37. What is the difference between Fact Tables and Dimension Tables?

- Fact Table: Stores measurable business transactions, such as sales, quantity and profit.
- Dimension Table: Stores descriptive information, such as customer name, product category and location.

38. What is cardinality in Power BI?

Answer: Cardinality defines how records in one table are related to records in another table.

Example: One-to-many and many-to-many are types of cardinality.

39. What is cross-filter direction?

Cross-filter direction determines how filters flow between related tables in a data model.

40. What is the difference between single-direction and both-direction filtering?

- Single Direction: Filters flow from one table to another.
- Both Direction: Filters can flow in both directions between related tables.

4. DAX (Data Analysis Expressions)

41. What is DAX in Power BI?

DAX stands for Data Analysis Expressions. It is a formula language used to create calculations, measures and calculated columns in Power BI.

Example:

```
None
Total Sales = SUM(Sales[Amount])
```

42. What is the difference between a calculated column and a measure?

- Calculated Column: Creates a new column by calculating a value for every row. The result is stored in the data model.
- Measure: Performs calculations dynamically based on the filters and context of the report.

43. What is the difference between SUM() and SUMX()?

- SUM() adds all values in a single column.
- SUMX() evaluates an expression for each row and then adds the results.

Example:

```
Total Sales = SUM(Sales[Amount])
```

```
Total Revenue =
SUMX(
    Sales,
    Sales[Quantity] * Sales[Price]
)
```

44. What is the purpose of the CALCULATE() function?

CALCULATE() evaluates an expression after modifying the filter context.

```
High Sales =
CALCULATE(
    SUM(Sales[Amount]),
    Sales[Amount] > 1000
)
```

45. What is the difference between COUNT() and COUNTA()?

- COUNT() counts non-blank numeric values in a column.
- COUNTA() counts non-blank values, including text and numbers.

46. What is the difference between COUNT() and DISTINCTCOUNT()?

- COUNT() counts non-blank numeric values, including duplicates.
- DISTINCTCOUNT() counts unique values.

Example: If the CustomerID column contains 101, 102, 101, 103, DISTINCTCOUNT returns 3.

47. What is the purpose of FILTER()?

FILTER() returns a table containing only the rows that satisfy a given condition.

Example:

```
None
High Sales =
FILTER(
    Sales,
    Sales[Amount] > 1000

)
```

48. What is the purpose of ALL()?

ALL() removes filters from a specified table or column. It is commonly used to calculate totals without the current filter restrictions.

Example:

```
None
Total All Sales =
CALCULATE(
    SUM(Sales[Amount]),
    ALL(Sales)

)
```

49. What is the difference between ALL() and ALLEXCEPT()?

- ALL() removes filters from the specified table or column.
- ALLEXCEPT() removes all filters from a table except the filters on specified columns.

50. What is the purpose of RELATED()?

RELATED() retrieves a value from a related table using an existing relationship.

Example:

```
None
Product Category =

RELATED(Products[Category])
```

51. What is the purpose of RELATEDTABLE()?

RELATEDTABLE() returns the rows from a related table that are connected to the current row.

52. What is the difference between IF() and SWITCH()?

- IF() checks a condition and returns one result if true and another if false.
- SWITCH() checks multiple possible values or conditions and returns the matching result.

Example:

```
None
Sales Status =
IF(
    Sales[Amount] > 1000,
    "High",
    "Low"

)
```

53. What is the purpose of DIVIDE()?

DIVIDE() performs division safely and allows us to specify an alternative result when the denominator is zero or blank.

Example:

```
Profit Margin =
DIVIDE(
    SUM(Sales[Profit]),
    SUM(Sales[Amount]),
    0

)
```

54. What are row context and filter context?

- Row Context: Refers to the current row being evaluated in a calculation.
- Filter Context: Refers to the filters applied to a calculation through slicers, visuals or DAX expressions.

55. What is context transition in DAX?

Context transition is the process of converting row context into filter context. It commonly happens when CALCULATE() is used inside a row context.

56. What is the purpose of SELECTEDVALUE()?

SELECTEDVALUE() returns the selected value from a column when only one distinct value is selected. Otherwise, it returns an alternative result if provided.

Example:

```
Selected Category =
SELECTEDVALUE(
    Products[Category],
    "Multiple Categories"

)
```

57. What is the difference between SUM() and AVERAGE()?

- SUM() calculates the total of numeric values.
- AVERAGE() calculates the arithmetic mean of numeric values.

58. What are time intelligence functions in DAX?

Time intelligence functions are used to perform calculations based on dates and time periods.

Examples:

- TOTALYTD()
- SAMEPERIODLASTYEAR()
- DATEADD()
- DATESMTD()
- DATESQTD()

59. What is the difference between TOTALYTD() and DATESYTD()?

- TOTALYTD() calculates the year-to-date total of an expression.
- DATESYTD() returns a table of dates from the beginning of the year to the current date in the filter context.

60. How do you calculate Year-over-Year (YoY) growth using DAX?

YoY growth compares the current year's value with the previous year's value.

Example:

```
None
115. Current Sales =

SUM(Sales[Amount])
```

```
None
116. Previous Year Sales =
```

```
CALCULATE(
    [Current Sales],
    SAMEPERIODLASTYEAR('Date'[Date])
)
```

None

```
YoY Growth =
DIVIDE(
    [Current Sales] - [Previous Year Sales],
    [Previous Year Sales],
    0
)
```

5. Power BI Visualisations

61. What are the different types of visualisations available in Power BI?

Power BI provides different visualisations to represent data, including:

- Bar Chart
- Column Chart
- Line Chart
- Pie Chart
- Donut Chart
- Area Chart
- Table
- Matrix
- Card
- KPI
- Map

- Scatter Chart
- Ribbon Chart

62. What is the difference between a Bar Chart and a Column Chart?

- Bar Chart displays data using horizontal bars.
- Column Chart displays data using vertical bars.

63. What is the difference between a Line Chart and an Area Chart?

- Line Chart displays trends using lines.
- Area Chart displays trends using lines with the area below them filled.

Both are useful for analysing changes over time.

64. What is a Stacked Column Chart?

A Stacked Column Chart displays multiple data categories within a single column to show the total and the contribution of each category.

65. What is a Combo Chart?

A Combo Chart combines two types of charts, usually a Column Chart and a Line Chart, to compare different measures.

Example: Monthly Sales using columns and Profit Margin using a line.

66. What is a Card visual?

A Card visual displays a single important numerical value.

Example: Total Revenue = ₹5,00,000.

67. What is a KPI visual?

A KPI visual displays the progress of a value towards a target.

Example: Actual Sales = ₹8 lakh, Target Sales = ₹10 lakh.

68. What is a Slicer?

A Slicer is an interactive visual that allows users to filter report data by selecting values.

Example: Selecting a particular year, region or product category.

69. What is the difference between a Slicer and a Filter?

- Slicer is a visible interactive visual on the report page.
- Filter is applied through the Filters pane or other filtering mechanisms and can work at visual, page or report level.

70. What is a Matrix visual?

A Matrix visual displays data in rows and columns with support for grouping, hierarchies and subtotals.

71. What is the difference between a Table and a Matrix?

- Table displays data in a simple row-and-column format.
- Matrix supports hierarchical grouping, expandable rows and subtotals.

72. What is Drill Down in Power BI?

Drill Down allows users to move from a higher level of data to a more detailed level within the same visual.

Example: Year → Quarter → Month → Day.

Drill Through allows users to navigate from one report page to another page containing detailed information about a selected data point.

Example: Selecting a product in a sales report to open a detailed product sales page.

74. What is Conditional Formatting?

Conditional Formatting changes the appearance of visual elements based on specific conditions or values.

Example: Displaying profit in green and loss in red.

75. What are Tooltips in Power BI?

Tooltips display additional information when a user hovers over a data point in a visual.

76. What is a Decomposition Tree?

A Decomposition Tree is an AI-powered visual that helps break down a measure into different categories to understand the factors influencing a result.

Example: Analysing total sales by region, product and salesperson.

77. What is a Key Influencers visual?

Key Influencers is an AI-powered visual that identifies factors that influence a particular outcome.

Example: Identifying which factors influence customer satisfaction.

78. What is a Ribbon Chart?

A Ribbon Chart displays changes in the ranking of categories over time.

Example: Comparing which product category has the highest sales each month.

79. What is a Map visual?

A Map visual represents geographical information using locations such as cities, states and countries.

Example: Displaying sales across different Indian states.

80. What is the difference between Map, Filled Map and Azure Maps?

- Map: Displays geographical locations using points or bubbles.
- Filled Map: Uses colours to highlight geographical regions based on values.
- Azure Maps: Provides advanced mapping features, including multiple map layers and richer geographical visualisations.

6. Power BI Service and Deployment

81. What is Power BI Service?

Power BI Service is Microsoft's cloud-based platform used to publish, share, manage and access Power BI reports and dashboards.

82. What is a Workspace in Power BI?

A Workspace is a collaborative environment where users can create, manage and share Power BI reports, dashboards and semantic models.

83. What is the difference between My Workspace and a shared Workspace?

Answer:

- My Workspace: A personal workspace used to store and manage your own Power BI content.
- Shared Workspace: A workspace where multiple users can collaborate and manage reports.

84. What is a Gateway in Power BI?

A Gateway is software that allows Power BI Service to securely connect to data sources located on local computers or company servers.

Example: Connecting an online Power BI report to a MySQL database running on a local server.

85. What is Scheduled Refresh?

Scheduled Refresh automatically updates the data in a Power BI semantic model at predefined times.

Example: Refreshing sales data every morning at 8 AM.

86. What is Row-Level Security (RLS)?

Row-Level Security restricts users from viewing certain rows of data based on their assigned roles or permissions.

Example: A Kerala sales manager can see only Kerala sales data.

87. What is the difference between Static RLS and Dynamic RLS?

- Static RLS: Uses fixed conditions to restrict data.
- Dynamic RLS: Uses the logged-in user's identity to determine which data they can access.

Example: Static RLS allows access to Kerala data, while Dynamic RLS identifies the logged-in user and displays the region assigned to that user.

88. What is a Power BI App?

A Power BI App is a packaged collection of reports and dashboards that can be distributed to users within an organisation.

89. What is the purpose of publishing a report?

Publishing uploads a report created in Power BI Desktop to Power BI Service, where it can be managed, shared and accessed by authorised users.

90. What are the different workspace roles in Power BI?

Power BI provides four main workspace roles:

- Admin – Can manage the workspace and its permissions.
- Member – Can create, edit and publish content.
- Contributor – Can create and edit content.
- Viewer – Can view workspace content.